

Cary Jones

Software Development Engineer | Embedded Systems & Satellite Platforms

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SUMMARY

Software Development Engineer with over 15 years of experience in developing high-performance embedded software solutions and distributed systems. Expert in **Python**, **Java**, and **Linux** environments, with a strong focus on improving software reliability and performance. Proven track record in troubleshooting complex production issues and implementing robust, scalable solutions. Experienced in Agile/Scrum methodologies and cross-functional team collaborations.

CORE COMPETENCIES & TECHNICAL SKILLS

- Programming Languages:** Python, Java, C++, Bash/Shell Scripting
- Embedded Systems:** OTA update systems, satellite platforms, IoT
- Development Environments:** Linux, Docker, Kubernetes, CI/CD
- Software Testing:** Automated testing frameworks, regression testing, root cause analysis
- Agile Methodologies:** Agile/Scrum, cross-functional team collaboration
- Problem Solving:** Debugging, performance optimization, system reliability

ACCOMPLISHMENTS

- Developed and deployed embedded software components for OTA update platforms, enhancing system reliability and performance.
- Engineered automated testing frameworks that improved software quality and reduced regression defects by **40%**.
- Led root cause analysis and implemented corrective actions that resolved complex production issues, reducing downtime by **30%**.
- Collaborated with cross-functional teams to design and deploy scalable solutions in Agile environments, accelerating development cycles by **25%**.

PROFESSIONAL EXPERIENCE

Principal Software Engineering Consultant / Technical Lead

Jan 2023 – Present | Los Angeles, CA / Remote

CGJSoftware, Inc.

- Operated corporate entity (CGJSoftware, Inc.) facilitating high-level B2B engineering contracts, advisory services, and cloud modernization for enterprise partners across healthcare, e-commerce, and data domains.
- Architected and deployed containerized Python microservices (FastAPI, Django, Sanic) on AWS (ECS, Fargate) and Docker, increasing system throughput by **45%** and reducing API latency to under **120ms**.
- Integrated LLMs and vector search capabilities (OpenAI API, LangChain) into client data pipelines, automating document classification and cutting manual data entry time by **60%**.
- Pioneered AI-augmented development practices across client teams using Cursor, Claude, and GitHub Copilot, slashing bug fix resolution times by **40%** and streamlining PR code review cycles.
- Delivered targeted legacy maintenance, cross-border software compliance reviews, and international cloud architecture strategies for remote enterprise clients.

Senior Software Engineer

May 2023 – July 2023 | CVS, Inc. (Remote)

- Analyzed and decoupled legacy monolithic codebases in preparation for domain-driven, microservices-oriented backend architecture.
 - Constructed containerized domain modules using Docker and Docker-Compose, improving localized integration testing speed by **50%**.
 - Advised engineering leadership on Domain-Driven Design (DDD) best practices, reducing inter-service dependencies across core application modules.
 - Executed proof-of-concept testing harness in Python to validate domain isolation without regressing critical business logic.
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Senior Software Engineer

October 2021 – November 2022 | Disney Corporation, Inc. (Remote)

- Engineered scalable backend modules for core enterprise financial and reporting systems processing over **\$10M+ in daily transaction metrics**.
 - Co-architected an automated accounting costing system, cutting end-of-month financial reconciliation processing times from **8 hours down to 45 minutes**.
 - Monitored and optimized cloud infrastructure on AWS, achieving **99.99% uptime** across critical reporting web applications.
 - Created high-performance Python database interfaces and Java unit test suites for PostgreSQL, ensuring **100% data integrity** during system migration.
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Python Software Developer and System Architect

July 2019 – October 2021 | CGJSoftware, Inc. (Remote)

- Served as primary architect and developer via S-Corp engagement, delivering customized cloud software and automated API architectures for mid-to-large business clients.
 - Containerized Python, React, and Node.js applications using Docker and deployed to GCP (Google Cloud Platform) using gcloud CLI, improving deployment frequency by **3x**.
 - Designed non-blocking asynchronous architectures utilizing AWS Pub/Sub messaging and Boto3 SDK, preventing application bottlenecks and ensuring **zero data loss** during peak loads.
 - Optimized complex SQL queries for AWS Redshift stress tests, increasing data warehouse reporting query performance by **38%**.
 - Created technical routing diagrams, system sequence charts, and detailed site maps to guide frontend engineering teams through complex application flows.
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API Developer, System Architect, DevOps Engineer, and Scrum Master

January 2018 – July 2019 | Alan Health & Science Corp. (Remote)

- Architected secure Python backend REST APIs for HIPAA-compliant healthcare platforms, servicing over **500k active user profiles**.
 - Managed a cross-functional offshore development team as Scrum Master, conducting PR reviews and establishing automated CI/CD pre-deploy scripts to eliminate code conflicts.
 - Configured production Linux Ubuntu environments, domain DNS (TXT, CNAME, MX), SSL/TLS security, and automated crontab backup procedures with **99.9% system availability**.
 - Authored comprehensive Git branching/rebasing documentation and trained junior engineers, reducing merge conflict incidents by **70%**.
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API Developer, Web Developer, and System Architect

July 2016 – January 2018 | CT-REI, Inc., Austin, TX

- Architected and built real estate investment evaluation and financial tracking web applications, growing organic web traffic by **45%** via technical SEO.
- Provisioned and managed Nginx web servers and MongoDB clusters on Google Cloud Compute, maintaining **sub-200ms load times** for end-users.
- Developed custom Python web scrapers to ingest appraisal district records, automating property offer calculations and boosting investment analysis throughput by **400%**.

Quality Assurance Engineer

Sept 2016 – Oct 2016 | Headway-Services, Austin, TX

- Designed comprehensive test plans and converted manual regression tests into fully automated test suites.
- Provided Agile/Scrum process consultation to streamline quality assurance workflows.
- Built an automated test framework in Python using unittest, Requests, and Nosetests to interface with Xojo desktop apps via HTTP, improving test execution speed by **65%**.
- Created custom XML parsing scripts to fix CDATA enclosure errors and transformed raw XML into clean HTML email templates using XSLT.

Python Programmer

May 2016 – July 2016 | Dell, Inc., Austin, TX

- Refactored legacy automated test framework software in Python to support enterprise server hardware testing.
- Eliminated circular dependencies and duplicated code across the project, reducing overall code bloat by **25%**.
- Refactored core ConfigParser modules into object-oriented models, reducing framework crash rates during long-running automated test executions by **80%**.

Software Developer in Test

October 2015 – May 2016 | Rackspace, Inc., Austin, TX

- Authored automated test protocols for ticket routing engines and customer-facing RESTful ticketing APIs.
- Developed an object-oriented Python REST client that executed **1,000+ automated test cases per hour** against core ticket processing services.
- Designed a dynamic test-generator using Python introspection to evaluate ticketing rulesets, uncovering critical edge-case bugs prior to production releases.

Software Developer and Scrum Master

April 2014 – August 2015 | KeyIngredient, Austin, TX

- Guided cross-functional Scrum teams, conducted peer code reviews, and implemented core e-commerce backend API endpoints on Google AppEngine.
- Automated B2B customer order fulfillment scripts compliant with ASC X12 EDI standards, cutting manual processing time by **90%**.
- Extracted and analyzed Google AppEngine datastore metrics to uncover consumer spending trends, driving a **15% increase in seasonal email campaign ROI**.

Python API Developer

August 2013 – April 2014 | Fidelity Investments, Irving, TX

- Supervised junior developers while building backend API services using Django REST Framework for datacenter network device automation.
 - Architected and delivered a web API that automated network device configuration, reducing manual provisioning errors by **95%** and setup time from **4 hours to 10 minutes**.
 - Mentored junior team members in Python best practices, API contract design, and Agile story estimation.
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Software Developer — QA Automation Engineer

January 2011 – June 2012 | FileTrek, Inc., Los Angeles, CA

- Developed VBA macros and COM/Office interop scripts to automate Microsoft Excel and Office workflows for enterprise file-management operations.
- Built programmatic integrations between desktop Office applications and backend systems, extracting structured data and reducing manual document processing.
- Debugged and maintained automation scripts across macro-enabled workbooks and COM-dependent Office components on Windows.

Quality Assurance & Network Integration Engineer

February 2005 – December 2009 | Hospira, Inc., San Diego, CA

- Verified and validated software for **Symbiq** and **Plum A+** hospital infusion pumps — requirements-based manual testing, smoke testing, formal verification, and defect tracking in regulated medical device workflows.
- Collaborated with firmware and hardware engineers to test embedded device behavior, troubleshoot software upgrades, and resolve complex integration issues across pump, server, and hospital network environments.
- Supported embedded systems over wired and wireless LANs; built recovery scripts and tools to restore infusion device software and configuration after firmware upgrade failures.
- Worked with **CAN bus** and embedded communication architectures; developed a .NET Telnet client automating host-to-device communication for verification and field-support scenarios.
- Installed and validated drug library management client/server software for networked infusion devices; served as verification team technical lead for device and PC networking issues.
- Built enterprise wireless security test architecture (EAP, Cisco ACS, FreeRadius) and delivered **5 formal verification run-throughs** for first-lot-to-stock ahead of schedule.

Field Wireman & Communications Specialist

September 1995 – September 1999 | United States Marine Corps, San Diego, CA & Overseas

- Configured and deployed tactical field telecommunications and wire networks under high-pressure military operations, achieving **100% mission communication uptime**.
- Awarded early promotion to Private First Class and Non-Commissioned Officer leadership responsibilities for exceptional accountability and performance.

EDUCATION & CERTIFICATIONS

Jun 2012 – Aug 2013: Full-time student at ITT Technical Institute, completing Associate of Science degree requirements between professional roles at FileTrek, Inc. and Fidelity Investments.

Associate of Science, Computer Electronics

2010 – 2013 | ITT Technical Institute, San Diego, CA

- Associate of Science Degree in Computer and Electronics Engineering Technology
- GPA 3.67

Basic Military Combat Skills

September 1995 | Marine Corps Recruit Depot, San Diego, CA

- Graduated from Basic Training; promoted to Private First Class